

deep implementation

Plan: EcoNexus Deep Specification Implementation

This plan transforms the existing 19-material, basic-compliance platform into a production-ready **847-SKU, fully-compliant European circular economy marketplace** with sophisticated agent negotiation, regulatory automation, and comprehensive material intelligence.

Foundation: The agent architecture, negotiation flow, and multi-party coordination are well-developed. We'll expand material taxonomy, implement compliance frameworks, enhance pricing mechanisms, and create formal data structures.

PHASE 1: Material Taxonomy & Classification System

Priority: CRITICAL | **Duration:** 3 weeks

1.1 Extended Material Database (847 SKUs)

Create: src/lib/constants/materials/

New files:

- [ferrous-metals.ts](src/lib/constants/materials/ferrous-metals.ts) - Class 1: 68 SKUs (HMS 1/2, shredded, stampings, turnings, coated steel, stainless grades, cast iron)
- [non-ferrous-metals.ts](src/lib/constants/materials/non-ferrous-metals.ts) - Class 2: 94 SKUs (Al alloys 6xxx/A380/A356, Cu grades, brass, zinc, precious metal scrap)
- [polymers.ts](src/lib/constants/materials/polymers.ts) - Class 3: 112 SKUs (PP/ABS/HDPE/PA66/POM/PET by grade, engineering plastics)

ng plastics, contaminated streams)

- [process-chemicals.ts](src/lib/constants/materials/process-chemicals.ts) - Class 4: 87 SKUs (cutting fluids, solvents, plating baths, acids)
- [organics.ts](src/lib/constants/materials/organics.ts) - Class 5: 42 SKUs (food processing waste, FOG, wastewater by BOD/COD)
- [energy-streams.ts](src/lib/constants/materials/energy-streams.ts) - Class 6: 28 SKUs (waste heat by temp grade, biogas, biomethane)
- [composites.ts](src/lib/constants/materials/composites.ts) - Class 7: 156 SKUs (multi-layer films, coated materials, assemblies)
- [specialized.ts](src/lib/constants/materials/specialized.ts) - Classes 8-12: 260 SKUs (textiles, ceramics, batteries, WEEE, construction materials)

****Master index:****

- [index.ts](src/lib/constants/materials/index.ts) - Exports all 847 materials, provides `getAllMaterials()`, `getMaterialByEWC()`, `searchMaterialsByClass()`

****Data structure per SKU:****

```typescript

```
interface MaterialSKU {
 id: string; // e.g. "1.1.3-shredded-scrap"
 class: 1-12;
 subclass: string;
 ewcCode: string;
 ewcDescription: string;
 tradeName: string;
 specifications: {
 composition: Record<string, Range>; // e.g. {Fe: {min: 85, max: 98}}
 physicalForm: string[];
 particleSize?: Range;
 }
}
```

```

 bulkDensity?: Range;
 };
 wasteTier: 'T1' | 'T2' | 'T3' | 'T4';
 hazard?: {
 unNumber?: string;
 adrClass?: string;
 packingGroup?: 'I' | 'II' | 'III';
 hProperties: string[]; // H1-H15
 };
 pricing: {
 basePrice: number; // EUR per unit
 unit: 't' | 'kg' | 'L' | 'm3' | 'kWh' | 'Nm3';
 priceRange: {min: number; max: number};
 marketIndex?: string; // LME, TSI, ICIS, etc.
 };
 qualityTiers: Array<{
 tier: 'Q1' | 'Q2' | 'Q3' | 'Q4';
 requirements: string[];
 priceMultiplier: number;
 contaminationMax: number; // percentage
 }>;
 processing: {
 recyclingRoute: string[];
 specialRequirements: string[];
 };
}
...

```

**\*\*Decision:\*\*** Structure as separate files by class to maintain navigability (847 SKUs in one file = unmaintainable). Use consistent naming: `{className}.{subclassNumber}.{itemNumber}-kebab-name`.

### ### 1.2 EWC Code Database

**\*\*Create:\*\*** [src/lib/constants/compliance/ewc-codes.ts](src/l

```
ib/constants/compliance/ewc-codes.ts)
```

```
38 relevant EWC code groups (spec mentions) expanded to *
*~180 specific codes** for manufacturing context:
```

- 02 (agriculture/food) - 8 codes
- 10 (thermal processes) - 24 codes
- 11 (chemical surface treatment) - 18 codes
- 12 (mechanical processing/metalworking) - 32 codes ★ Primary
- 13 (oil wastes) - 14 codes
- 14 (organic solvents) - 12 codes
- 15 (packaging) - 8 codes
- 16 (WEEE) - 16 codes
- 17 (construction) - 12 codes
- 19 (waste treatment facilities) - 22 codes
- Others - 14 codes

```
Data structure:
```

```
```typescript
```

```
interface EWCCode {  
  code: string; // "12 01 01"  
  description: string;  
  hazardous: boolean; // asterisk in code  
  wasteTier: 'T2' | 'T3' | 'T4';  
  hProperties?: string[];  
  restrictions: {  
    requiresLicense: boolean;  
    requiresNotification: boolean;  
    crossBorderRules: 'green' | 'amber' | 'red';  
  };  
  typicalSources: string[];  
  disposalMethods: string[];  
  recoveryOperations: string[]; // R1-R13  
}  
```
```

**\*\*Helper functions:\*\***

- `validateEWCCode(code: string)` - format check + exists
- `getEWCProperties(code: string)` - returns full record
- `isHazardous(code: string)` - checks asterisk
- `getWasteTier(code: string)` - T1-T4 classification

### ### 1.3 Contamination & Quality Assessment

**\*\*Enhance:\*\*** [src/lib/material-intelligence/scoring.ts](src/lib/material-intelligence/scoring.ts)

**\*\*Add functions:\*\***

- `calculateContaminationLevel(composition, baseline): 'C0' | 'C1' | 'C2' | 'C3'` - based on spec thresholds (<0.5%, 0.5-3%, >3%)
- `determineQualityTier(material, analysis, certifications): 'Q1' | 'Q2' | 'Q3' | 'Q4'` - implements 4-tier certification logic
- `calculatePriceImpact(qualityTier, contaminationLevel): number` - multiplier from spec (Q1: +18-25%, C3: -30-60%)
- `assessProcessability(material): ProcessabilityScore` - difficulty, routes, requirements

---

**## \*\*PHASE 2: Compliance & Regulatory Framework\*\***

**\*Priority:** CRITICAL (Legal requirement) | **Duration:** 2 weeks\*

### ### 2.1 Regulatory Constants

**\*\*Create:\*\*** [src/lib/constants/compliance/](src/lib/constants/compliance/)

**\*\*Files:\*\***

- [adr-classifications.ts](src/lib/constants/compliance/adr-classifications.ts) - ADR 2023 dangerous goods (Classes 1-9, p

- [reach-substances.ts](src/lib/constants/compliance/reach-substances.ts) - SVHC list (240+ substances of very high concern with CAS numbers, concentration limits)
- [waste-shipment-rules.ts](src/lib/constants/compliance/waste-shipment-rules.ts) - WSR 1013/2006 green/amber/red list logic
- [csrd-mappings.ts](src/lib/constants/compliance/csrd-mappings.ts) - ESRs E1/E5 category mappings, Scope 3 calculation formulas
- [regulatory-config.ts](src/lib/constants/compliance/regulatory-config.ts) - Master export + helper functions

### ### 2.2 Compliance Validation Engine

**\*\*Create:\*\*** [src/lib/compliance/validation-engine.ts](src/lib/compliance/validation-engine.ts)

**\*\*Core functions (from spec logic blocks):\*\***

```
```typescript
// From S10 automated compliance gates
function preDealComplianceCheck(deal: Deal, material: MaterialPassport): ComplianceResult {
  // 1. T4 hazmat check
  // 2. Cross-border amber/red check
  // 3. REACH SVHC disclosure
  // 4. WEEE authorization
  // 5. CSRD tagging
  // Returns: {passed: boolean, flags: string[], blockers: string[]}
}

function validateTransportCompliance(material, carrier, route): TransportCompliance {
  // ADR cert validation
```

```

    // UN number matching
    // Packing group check
    // Emergency info requirements
}

function assessCrossBorderRules(material, originCountry, dest
Country): ShipmentRules {
    // Green/amber/red classification
    // Notification requirements
    // Consent periods
    // Financial guarantee needs
}
...

```

2.3 Database Schema Extensions

```

**Modify:** [src/types/database.ts](src/types/database.ts)

**Add to `waste_streams` table:**
```typescript
ewc_code: string;
waste_tier: 'T1' | 'T2' | 'T3' | 'T4';
hazard_properties: string[]; // H1-H15
un_number?: string;
adr_class?: string;
packing_group?: string;
reach_svhc_flag: boolean;
svhc_substances?: Array<{cas: string; concentration: number}>
>;
...

Add to `material_passports` table:
```typescript
ewc_code: string;
contamination_level: 'C0' | 'C1' | 'C2' | 'C3';
batch_reference: string;

```

```

source_process: string;
generation_date: Date;
available_from: Date;
monthly_forecast: number[]; // 12 months
composition: Record<string, number>; // element percentages
moisture_pct?: number;
particle_size_mm?: {min: number; max: number};
bulk_density?: number;
hazard_codes: string[];
certifications: string[];
carbon_method?: string;
carbon_avoided?: number;
...

```

****New tables:****

```

```typescript
// Company certifications & licenses
company_certifications {
 id, company_id, certification_type,
 certification_number, issuing_authority,
 valid_from, valid_until,
 scope, document_url
}

// Material market pricing (time-series)
material_market_prices {
 id, material_id, price_source,
 price_eur, unit, recorded_at,
 region, grade
}

// Audit trail (comprehensive)
audit_log {
 id, timestamp, actor_type, actor_id,
 action_type, entity_type, entity_id,

```



```

 changes_json, ip_address, user_agent
}

// Compliance checks log
compliance_checks {
 id, deal_id, check_type,
 check_timestamp, result,
 flags_json, blockers_json,
 override_by?, override_reason?
}
...

PHASE 3: Enhanced Agent Logic & Negotiation
Priority: HIGH | Duration: 3 weeks

3.1 ZOPA Calculation Engine

Create: src/lib/agents/zopa-calculator.ts

Implements spec section S5:

```typescript
interface ZOPACalculation {
    reservationPrice: number; // Floor (seller) or ceiling (buyer)
    targetPrice: number; // Desired outcome
    openingPosition: number; // First offer
    batna: number; // Best alternative
    batnaDescription: string;
    walkawayPoint: number;
}

function calculateSellerZOPA(

```

```

    material: MaterialSKU,
    wasteStream: WasteStream,
    marketData: PriceData,
    currentDisposalCost: number
): ZOPACalculation {
    // Reservation = disposal cost avoided + min margin
    // Target = market rate × 1.05
    // Opening = target × 1.15
    // BATNA = current buyer or spot market rate
}

function calculateBuyerZOPA(
    material: MaterialSKU,
    needDetails: MaterialNeed,
    marketData: PriceData,
    virginAlternative: number
): ZOPACalculation {
    // Reservation = virgin material cost or current supplier
    // Target = market rate × 0.92
    // Opening = target × 0.88
    // BATNA = current supplier price
}

function checkZOPAOverlap(
    sellerZOPA: ZOPACalculation,
    buyerZOPA: ZOPACalculation
): {exists: boolean; range?: [number, number]; gap?: number}
{
    // If seller.reservation ≤ buyer.reservation: overlap exists
    // Range = [seller.reservation, buyer.reservation]
    // If no overlap, return gap size for diagnostics
}
...

```

3.2 Enhanced Negotiation State Machine

```
**Enhance:** [src/lib/agents/agent-runner.ts](src/lib/agents/agent-runner.ts)
```

```
**Add 7-state machine (from spec S2):**
```

```
```typescript
```

```
enum NegotiationState {
 MONITORING = 'monitoring',
 EVALUATING = 'evaluating',
 CONTACTING = 'contacting',
 NEGOTIATING = 'negotiating',
 PROPOSING = 'proposing',
 AWAITING_APPROVAL = 'awaiting',
 EXECUTING = 'executing'
}
```

```
// State transitions with gates
```

```
function transitionState(
 currentState: NegotiationState,
 trigger: string,
 context: NegotiationContext
): NegotiationState {
 // Implements state machine from spec with human approval gates
}
```
```

```
**Expand to 5-round negotiation (from 3):**
```

- Round 1: Opening (anchor phase)
- Round 2: Conditional concession (if X then Y)
- Round 3: Volume/terms trade (multi-dimensional)
- Round 4: Final positions (Fibonacci pattern)
- Round 5: Resolution or deadlock escalation

```
**Add Fibonacci concession pattern:**
```

```

```typescript
function calculateConcession(round: number, totalGap: number): number {
 const pattern = [0.50, 0.25, 0.12, 0.06]; // % of gap to close
 return totalGap * (pattern[round - 2] || 0.03);
}
```

```

3.3 Enhanced Scoring Algorithm

```

**Enhance:** [src/lib/agents/scoring-engine.ts](src/lib/agents/scoring-engine.ts)

```

```

**Implement spec S2 scoring weights:**

```

```

```typescript
function scoreOpportunity(opportunity: Opportunity): OpportunityScore {
 // Spec: Price delta (30pts) + Volume (20pts) + Distance (15pts) +
 // Reputation (15pts) + Quality (10pts) + Contract term (5pts) +
 // Carbon bonus (5pts) = 100pts max

 const priceScore = calculatePriceDelta(/* ... */); // 0-30
 const volumeScore = calculateVolumeMatch(/* ... */); // 0-20
 const distanceScore = calculateDistanceScore(/* ... */); // 0-15
 const reputationScore = opportunity.counterparty.rating * 3; // 0-15
 const qualityScore = calculateQualityAlignment(/* ... */); // 0-10
 const termScore = calculateTermLength(/* ... */); // 0-5
 const carbonBonus = Math.min(5, opportunity.carbonSaved / 1

```

```
00); // 0-5
```

```
 return {
 total: sum(all scores),
 breakdown: {price, volume, distance...},
 threshold: total < 45 ? 'IGNORE' :
 total < 60 ? 'MONITOR' :
 total < 75 ? 'CONTACT' :
 total < 90 ? 'PRIORITIZE' : 'URGENT'
 };
 }
 ...
}
```

### ### 3.4 Constraint Enforcement

```
Create: src/lib/agents/constraint-checker.ts
```

```
Implements spec S2 constraint logic:
```

```
```typescript
function checkConstraints(
  opportunity: Opportunity,
  agentConstraints: AgentConstraints
): ConstraintResult {
  const hardFails = [];
  const softFlags = [];
  let score = opportunity.rawScore;

  // Hard constraints (reject if ANY fail)
  if (opportunity.distance > agentConstraints.max_distance)
    hardFails.push('distance');
  if (opportunity.material.wasteTier === 'T4' &&
    !opportunity.counterparty.hazmat_certified)
    hardFails.push('hazmat');
  // ... (all hard constraints from spec)
```

```

    // Soft constraints (reduce score)
    if (opportunity.price < agentConstraints.target_price * 0.8
5)
        score -= 20;
    // ... (all soft constraints from spec)

    return {
        passed: hardFails.length === 0,
        hardFails,
        softFlags,
        adjustedScore: score
    };
}
...

```

```

## **PHASE 4: Digital Passport Enhancement**
*Priority: MEDIUM | Duration: 2 weeks*

```

```

### 4.1 Complete Passport Schema

```

```

**Enhance:** [src/lib/material-intelligence/passport-service.
ts](src/lib/material-intelligence/passport-service.ts)

```

```

**Add all fields from spec S6:**

```

```

```typescript
interface CompletePassport {
 // Identity (existing)
 passport_id, batch_reference, created_at, version,

 // Material Identity (enhanced)
 material_class: 1-12,
 material_subclass: string,

```

```
ewc_code: string,
cas_number?: string,
trade_name: string,

// Source (new)
source_process: string,
source_location: GeoPoint,
generation_date: Date,
available_from: Date,

// Quantity (enhanced)
monthly_forecast: number[12],

// Composition (new)
composition: Map<element, percentage>,
contamination_level: C0-C3,
moisture_pct?: number,
particle_size_mm?: Range,
bulk_density?: number,

// Hazard (new)
waste_tier: T1-T4,
hazard_codes: string[],
un_number?: string,
adr_class?: string,
packing_group?: string,

// Quality (enhanced)
quality_tier: Q1-Q4,
certifications: string[],
analysis_docs: FileRef[],

// Environmental (enhanced)
carbon_footprint_kg?: number,
carbon_method?: string,
carbon_avoided?: number,
```

```

 // Chain of custody (enhanced)
 custody_chain: CustodyEvent[],
 current_holder: UUID,
 deal_reference?: UUID
 }
 ...

```

### ### 4.2 Custody Chain & Transfers

**\*\*Enhance:\*\*** Passport transfer tracking with immutable event log

**\*\*Add functions:\*\***

- `appendCustodyEvent(passportId, event)` - append-only, no updates
- `verifyChainIntegrity(passportId)` - check hash chain
- `exportProvenanceReport(passportId)` - full history PDF

---

**## \*\*PHASE 5: Human Control Layer\*\***

**\*Priority: HIGH | Duration: 2 weeks\***

### ### 5.1 Risk Flagging System

**\*\*Create:\*\*** [src/lib/agents/risk-assessor.ts](src/lib/agents/risk-assessor.ts)

**\*\*Implements spec S7 risk flags:\*\***

```

```typescript
enum RiskLevel { HIGH, MEDIUM, LOW }

```

```

interface RiskFlag {
  level: RiskLevel;
}

```



```

    category: string;
    message: string;
    blocking: boolean; // HIGH flags block deal
}

function assessDealRisk(deal: Deal, context: DealContext): RiskFlag[] {
    const flags: RiskFlag[] = [];

    // HIGH RISK (block deal) - 6 checks from spec
    if (!deal.counterparty.verified && deal.counterparty.isNew)
        flags.push({level: HIGH, category: 'counterparty',
                    message: 'Unverified new counterparty', blocking: true});

    if (deal.material.wasteTier === 'T4' &&
        !deal.carrier.hazmatCert)
        flags.push({level: HIGH, category: 'compliance',
                    message: 'T4 waste without hazmat certified carrier',
                    blocking: true});

    // ... all HIGH flags from spec

    // MEDIUM FLAGS (warn + proceed) - 6 checks
    if (deal.isFirstWithCounterparty)
        flags.push({level: MEDIUM, category: 'relationship',
                    message: 'First deal with this company', blocking: false});

    // LOW FLAGS (informational) - 6 checks

    return flags;
}
...

```

5.2 Approval Workflow

```
**Create:** [src/lib/workflows/approval-workflow.ts](src/lib/workflows/approval-workflow.ts)
```

```
**Implements spec S7 state machine:**
```

```
```typescript
```

```
enum ApprovalState {
 DEAL_PROPOSED = 'deal_proposed',
 NOTIFICATION_SENT = 'notification',
 UNDER_REVIEW = 'review',
 APPROVED = 'approved',
 REJECTED = 'rejected',
 EDIT_REQUESTED = 'edit'
}
```

```
function processApprovalDecision(
 dealId: string,
 decision: 'approve' | 'reject' | 'edit',
 humanId: string,
 edits?: Partial<Deal>
) : ApprovalResult {
 switch(decision) {
 case 'approve':
 // Single tap. Make binding. Generate contract. Notify
 counterparty.
 break;
 case 'edit':
 // Human adjusts one term. Agent re-negotiates that spe
 cific term.
 break;
 case 'reject':
 // Require reason. Update agent preference model (learn
 ing).
 break;
```

```
}
}
...
```

### ### 5.3 Emergency Controls

**\*\*Create:\*\*** [src/lib/agents/emergency-controls.ts](src/lib/agents/emergency-controls.ts)

**\*\*4 control levels from spec:\*\***

- `pauseNegotiations(agentId)` - freeze active, timeout gracefully
- `pauseMaterial(agentId, materialId)` - stop one material type
- `emergencyStop(agentId)` - complete halt, expire pending
- `undoLastProposal(dealId)` - within 2h, before counterparty views

### ### 5.4 Audit Trail

**\*\*Enhance:\*\*** [src/types/database.ts](src/types/database.ts) a udit\_log table

**\*\*Log every:\*\***

- Agent action (with content hash)
- Negotiation message (full text preserved)
- Human decision (approval/reject with mandatory reason)
- Constraint change (who, when, old→new)

**\*\*Retention:\*\*** 7 years (EU requirement), exportable anytime

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**## \*\*PHASE 6: Pricing Mechanisms & Market Intelligence\*\***

**\*Priority: MEDIUM | Duration: 2 weeks\***

### ### 6.1 Market Price Integration

**\*\*Create:\*\*** [src/lib/pricing/market-data-feeds.ts](src/lib/pricing/market-data-feeds.ts)

**\*\*Implements spec S8 price sources:\*\***

```
```typescript
interface PriceFeed {
  material: string;
  source: 'LME' | 'TSI' | 'ICIS' | 'OPIS' | 'EPEX' | 'TTF' | 'ICE';
  frequency: 'realtime' | 'hourly' | 'daily' | 'weekly';
  apiEndpoint?: string;
  transformFn: (raw: any) => number;
}

const PRICE_FEEDS: PriceFeed[] = [
  {material: 'steel-scrap-hms1', source: 'TSI', frequency: 'weekly', ...},
  {material: 'aluminum-*', source: 'LME', frequency: 'daily', ...},
  {material: 'copper-*', source: 'LME', frequency: 'realtime', ...},
  // ... all materials from spec table
];

async function fetchMarketPrice(materialId: string): Promise<PriceData> {
  // Lookup feed, call API, cache, return
}
```

Decision: Use API mocks initially (random walk from baseline). Phase 2: integrate real feeds (LME free tier, TSI subscription).
```

### ### 6.2 Dynamic Pricing Strategy

**\*\*Create:\*\*** [src/lib/agents/pricing-strategy.ts](src/lib/agents/pricing-strategy.ts)

**\*\*Agent learns from:\*\***

- Accepted vs. rejected deals
- Market price movements
- Seasonal patterns
- Counterparty behavior

```typescript

function adjustTargetPrice(
 agent: Agent,
 material: MaterialSKU,
 recentDeals: Deal[]
)

): number {

// If acceptance rate < 50% → agent is too aggressive → lower ask

// If acceptance rate > 80% → leaving money on table → raise ask

// Weighted by deal recency (exponential decay)

}

```

---

**## \*\*PHASE 7: UI Enhancements\*\***

**\*Priority: MEDIUM | Duration: 2 weeks\***

### ### 7.1 Material Flow Creation (Enhanced)

**\*\*Enhance:\*\*** [src/app/(dashboard)/materials/flow/create/page.tsx](src/app/(dashboard)/materials/flow/create/page.tsx)

#### **\*\*Changes:\*\***

- Material selector: Search 847 SKUs (autocomplete with class filter)
- EWC code: Auto-populate from selected SKU, allow override with validation
- Waste tier: Auto-assign from EWC, display badge (T1-T4)
- Hazmat section: Conditional render if T4, require UN number/ADR class
- Quality assessment: Contamination level (C0-C3) visual slider, attach lab reports for Q2+
- Composition: Dynamic element input (Fe, Cu, Zn...) with percentage, validates against SKU spec
- Monthly forecast: 12-month input graph

#### **### 7.2 Digital Passport Viewer**

**\*\*Create:\*\*** [src/app/(dashboard)/digital-passport/[id]/page.tsx](src/app/(dashboard)/digital-passport/[id]/page.tsx)

#### **\*\*Sections:\*\***

- **\*\*Identity:\*\*** QR code, passport ID, batch reference, version history
- **\*\*Material:\*\*** Class/subclass tree, EWC code badge, trade name
- **\*\*Source:\*\*** Company, process description, generation date, location map
- **\*\*Composition:\*\*** Element breakdown chart, contamination level badge, quality tier badge
- **\*\*Hazard:\*\*** Waste tier badge, H-properties list, ADR details, transport warnings
- **\*\*Certifications:\*\*** Quality tier badges, attached documents, verification status
- **\*\*Environmental:\*\*** Carbon footprint chart, avoided emissions, method
- **\*\*Custody Chain:\*\*** Timeline of transfers, current holder, deal references

### ### 7.3 Agent Dashboard

**\*\*Enhance:\*\*** [src/app/(dashboard)/agent/page.tsx](src/app/(dashboard)/agent/page.tsx)

**\*\*Add:\*\***

- Current state indicator (7-state machine visual)
- Active negotiations list with round number
- Pending approvals queue with risk flags (HIGH/MEDIUM/LOW badges)
- Emergency controls panel (pause/stop buttons)
- Constraint editor (all fields from spec S2 table)
- Performance metrics: acceptance rate, avg deal value, carbon saved
- Learning insights: "Agent adjusted target prices after 65% rejection rate"

### ### 7.4 Compliance Dashboard

**\*\*Create:\*\*** [src/app/(dashboard)/compliance/page.tsx](src/app/(dashboard)/compliance/page.tsx)

**\*\*Sections:\*\***

- **\*\*Certifications:\*\*** Company hazmat permits, WEEE authorization, ISO certs (expiry alerts)
- **\*\*Active Checks:\*\*** Recent compliance validations, flags, blockers
- **\*\*Cross-Border Queue:\*\*** Pending WSR notifications (30-day countdown timers)
- **\*\*CSRD Reporting:\*\*** Scope 3 data export, ESRS category breakdown, annual progress
- **\*\*Audit Log:\*\*** Searchable trail of all actions, filterable by entity/actor/date

---

## ## \*\*PHASE 8: Platform Revenue & Analytics\*\*

\*Priority: LOW (Future) | Duration: 1 week\*

### ### 8.1 Subscription Management

**\*\*Create:\*\*** [src/lib/billing/subscription-tiers.ts](src/lib/billing/subscription-tiers.ts)

**\*\*3 tiers from spec S8:\*\***

- Starter: €490/mo (1 agent, 5 streams, 50km)
- Growth: €1,200/mo (unlimited streams, 150km, API)
- Enterprise: €3,500/mo (multi-site, super agent, SLA)

**\*\*Integrate:\*\*** Stripe for payments, usage tracking, upgrade flows

### ### 8.2 Transaction Fees

**\*\*Add to deals table:\*\*** `platform\_fee\_eur` (calculated on approval: 0.8% standard, 1.5% multi-party, 3% logistics)

### ### 8.3 Market Intelligence Products

**\*\*Create:\*\*** [src/app/api/market-intelligence/](src/app/api/market-intelligence/)

- Regional price benchmarks API
- Material flow reports (anonymized)
- Trend analysis

---

## ## \*\*Implementation Sequence & Dependencies\*\*

### ### Week 1-3: Foundation (Phase 1 + Phase 2)

1. Material taxonomy (847 SKUs) - **\*\*No dependencies\*\***



2. EWC codes database - **\*\*Parallel with #1\*\***
3. Compliance constants (ADR, REACH, WSR, CSRD) - **\*\*Parallel with #1-2\*\***
4. Database schema extensions - **\*\*After #1-3 finalized\*\***
5. Contamination/quality assessment functions - **\*\*After #1\*\***

#### ### Week 4-6: Core Logic (Phase 3)

6. ZOPA calculator - **\*\*Requires Phase 1 (pricing data)\*\***
7. Enhanced negotiation (5-round + state machine) - **\*\*Requires #6\*\***
8. Enhanced scoring (spec weights) - **\*\*Requires Phase 1\*\***
9. Constraint checker - **\*\*Requires Phase 2 (compliance)\*\***
10. Compliance validation engine - **\*\*Requires Phase 2\*\***

#### ### Week 7-8: Data Enhancement (Phase 4)

11. Complete passport schema - **\*\*Requires Phase 1 + 2\*\***
12. Custody chain - **\*\*Requires #11\*\***

#### ### Week 9-10: Human Layer (Phase 5)

13. Risk assessor - **\*\*Requires Phase 3 + 4\*\***
14. Approval workflow - **\*\*Requires #13\*\***
15. Emergency controls - **\*\*Requires Phase 3\*\***
16. Enhanced audit trail - **\*\*Requires Phase 2 (schema)\*\***

#### ### Week 11-12: Market Data (Phase 6)

17. Market price feeds - **\*\*Requires Phase 1\*\***
18. Dynamic pricing strategy - **\*\*Requires Phase 3 + #17\*\***

#### ### Week 13-14: UI (Phase 7)

19. Enhanced material flow creation - **\*\*Requires Phase 1 + 2\*\***
20. Passport viewer - **\*\*Requires Phase 4\*\***
21. Agent dashboard - **\*\*Requires Phase 3 + 5\*\***
22. Compliance dashboard - **\*\*Requires Phase 2 + 5\*\***

#### ### Week 15+ : Revenue (Phase 8 - Future)

23. Subscription management - **\*\*Standalone\*\***  
24. Transaction fees - **\*\*Requires Phase 3 (deals)\*\***  
25. Market intelligence APIs - **\*\*Requires Phase 6\*\***

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## ## **\*\*Testing Strategy\*\***

### ### Unit Tests (Each Phase)

- Material lookup by EWC code
- Quality tier calculation
- Contamination level assessment
- ZOPA overlap detection
- Risk flag generation
- Compliance validation rules
- Price impact calculations

### ### Integration Tests

- End-to-end negotiation (5 rounds)
- Multi-party deal with compliance checks
- Passport creation with all fields
- State machine transitions
- Approval workflow (approve/reject/edit paths)

### ### Compliance Tests (Critical)

- **\*\*T4 hazmat blocking:\*\*** Material with wasteTier='T4' + no hazmat cert = deal blocked
- **\*\*Cross-border amber:\*\*** Origin=DE, Dest=FR, material=amber list = notification workflow triggered
- **\*\*REACH SVHC:\*\*** Material with SVHC >0.1% = disclosure required
- **\*\*WEEE authorization:\*\*** EWC 16 02 \* + no R4 auth = blocked
- All EWC codes validate (no typos)
- All UN numbers map to correct ADR classes

### ### Performance Tests

- 847 SKU search: <200ms
- Opportunity scoring: <50ms per opportunity
- Passport creation: <500ms
- Feed scan (1000 posts): <3s
- Compliance check: <100ms

### ### Manual QA Scenarios

1. **\*\*Create passport for each material class (1-12):\*\*** Verify all fields render correctly
2. **\*\*Trigger each risk flag:\*\*** HIGH (6 types), MEDIUM (6 types), LOW (6 types)
3. **\*\*Emergency stop test:\*\*** Stop agent mid-negotiation → verify graceful timeout with counterparty notification
4. **\*\*T4 hazmat deal:\*\*** Attempt deal without carrier ADR cert → verify hard block
5. **\*\*5-round negotiation:\*\*** Full negotiation from opening to resolution with Fibonacci concessions
6. **\*\*Multi-party approval:\*\*** 3-way deal where 1 party rejects → verify deal cancellation
7. **\*\*Audit trail:\*\*** Change constraint → verify log captures old/new values + user ID

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### ## \*\*Rollout Strategy\*\*

#### ### Phase 1 Rollout: Material Taxonomy (Week 3)

- **\*\*Deploy:\*\*** 847 SKU files to production
- **\*\*Communication:\*\*** Email existing users about expanded material options
- **\*\*Migration:\*\*** Map existing 19 materials to new SKU IDs
- **\*\*Validation:\*\*** Run script to ensure all waste streams have valid material IDs

#### ### Phase 2 Rollout: Compliance (Week 5)

- **\*\*Deploy:\*\*** EWC/ADR/REACH validators

- **\*\*Alert:\*\*** Flag existing waste streams missing EWC codes (grace period: 30 days)
- **\*\*UI:\*\*** Show compliance status badges on all materials
- **\*\*Training:\*\*** Webinar on EWC code selection + hazmat requirements

### ### Phase 3 Rollout: Enhanced Agents (Week 7)

- **\*\*Deploy:\*\*** ZOPA + 5-round negotiation (feature flag initially)
- **\*\*A/B Test:\*\*** 50% of agents use new logic, 50% old → compare deal success rate
- **\*\*Monitor:\*\*** Watch for negotiation deadlocks (if >20% increase, rollback)
- **\*\*Full rollout:\*\*** After 2 weeks of stable A/B

### ### Phase 4 Rollout: Passports (Week 9)

- **\*\*Deploy:\*\*** Enhanced passport schema
- **\*\*Migration:\*\*** Backfill existing passports with default values (C1 contamination, Q3 quality)
- **\*\*Feature:\*\*** "Upgrade to Q2" CTA for users to add lab reports

### ### Phase 5 Rollout: Human Controls (Week 11)

- **\*\*Deploy:\*\*** Risk flags + approval workflow
- **\*\*Default:\*\*** All deals require approval initially (auto-approve threshold = €0)
- **\*\*Education:\*\*** Dashboard tour explaining risk levels
- **\*\*Gradual:\*\*** Users increase auto-approve threshold at own pace

### ### Phase 6 Rollout: Market Data (Week 13)

- **\*\*Deploy:\*\*** Mock price feeds (Phase 1)
- **\*\*Announce:\*\*** Real feeds coming Q3 2026 (pending LME/TSI contracts)
- **\*\*Feature:\*\*** Show "Market price: €142/t (ICIS estimate)" on all materials

### ### Phase 7 Rollout: UI (Week 15)

- **\*\*Deploy:\*\*** Material flow creation (backward compatible)
- **\*\*Deploy:\*\*** Passport viewer (new page)
- **\*\*Deploy:\*\*** Agent dashboard (replaces old)
- **\*\*Deploy:\*\*** Compliance dashboard (new section)
- **\*\*Communication:\*\*** Product update email with video tutorials

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## ## **\*\*Key Decisions\*\***

### ### Database Architecture

**\*\*Decision:\*\*** Keep dual structure (Supabase + TypeScript types) initially. Phase 2: Generate Prisma schema from types, migrate incrementally.

**\*\*Rationale:\*\*** Avoids breaking existing Supabase setup. Prisma migration is large effort; defer to avoid blocking other phases.

### ### Material Data Storage

**\*\*Decision:\*\*** 847 SKUs as TypeScript constants (not DB).

**\*\*Rationale:\*\*** Static data, version-controlled, type-safe, no query overhead. When mature (Phase 2: 2027), migrate to DB with seeding for user-contributed materials.

### ### EWC Code Management

**\*\*Decision:\*\*** TypeScript constants initially, DB later for user additions.

**\*\*Rationale:\*\*** EU EWC list is stable (updated ~yearly). No need for runtime changes. Future: allow companies to propose new codes (admin approval queue).

### ### Compliance Enforcement

**\*\*Decision:\*\*** Pre-flight checks only (before deal proposal).

No post-facto validation.

**\*\*Rationale:\*\*** Block early, fail fast. Allowing non-compliant negotiations wastes agent time and risks legal exposure. Better to reject at evaluation stage.

### ### Agent Learning

**\*\*Decision:\*\*** Constraint auto-adjustment OFF by default. Human must enable per agent.

**\*\*Rationale:\*\*** Too risky to learn wrong patterns from small sample sizes. E.g., if first 5 deals at low prices succeed, agent might permanently lower targets. Require explicit opt-in + minimum sample size (50 deals).

### ### Pricing Data

**\*\*Decision:\*\*** Phase 1 = mock feeds (random walk from baseline). Phase 2 = real APIs.

**\*\*Rationale:\*\*** Real feeds require paid subscriptions (LME €2k/mo, TSI €5k/mo, ICIS €3k/mo). Not justifiable pre-revenue. Mock feeds enable testing pricing logic without cost.

### ### Multi-Tenancy

**\*\*Decision:\*\*** Current single-tenant per company. Super agents span multiple companies but data isolated.

**\*\*Rationale:\*\*** True multi-tenant with row-level security (RLS) is complex. Current architecture works for 100-500 companies. Revisit at scale (5,000+ companies).

### ### Negotiation Rounds

**\*\*Decision:\*\*** Expand from 3 to 5 rounds.

**\*\*Rationale:\*\*** Spec specifies 5-round protocol with distinct purposes per round. More rounds = higher convergence rate (spec claims 94% vs. 68% with 3 rounds). Minimal performance cost (<500ms per round).

### ### Quality Tiers

**\*\*Decision:\*\*** Keep 4-tier system (Q1-Q4), not expand to 5 or

6.

**\*\*Rationale:\*\*** Spec defines 4 tiers with clear certification requirements. More tiers = confusion + diminishing value. Market standard (ISO classifications) also use 3-4 tiers.

#### ### Waste Tiers

**\*\*Decision:\*\*** Use T1-T4 (not 0-VI from some EU docs).

**\*\*Rationale:\*\*** Spec consistently uses T1 (by-product), T2 (end-of-waste), T3 (non-hazardous waste), T4 (hazardous waste). Simpler than 7-tier system, maps cleanly to regulatory requirements.

#### ### Carbon Accounting

**\*\*Decision:\*\*** Passport stores avoided emissions, not full LCA.

**\*\*Rationale:\*\*** Full lifecycle assessment requires cradle-to-grave data (upstream not available). Focus on "avoided virgin production" baseline comparison (spec approach). Sufficient for CSRD Scope 3 Category 5 reporting.

#### ### Blockchain Integration

**\*\*Decision:\*\*** Deferred to Phase 9 (2027+).

**\*\*Rationale:\*\*** Spec mentions blockchain for passport immutability, but current append-only event log with hash chain provides 90% of benefit. Blockchain adds complexity (Ethereum gas fees, node management). Revisit when customer demand materializes.

#### ### API Rate Limits

**\*\*Decision:\*\*** LME/TSI/ICIS APIs = daily batch fetch (4 AM UTC), cache 24h.

**\*\*Rationale:\*\*** Real-time pricing unnecessary for waste streams (not commodities trading). Daily updates sufficient. Reduces API costs 24x.

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## ## \*\*Success Metrics\*\*

### ### Phase 1 (Material Taxonomy)

-  All 847 SKUs accessible via API
-  Search returns results in <200ms
-  100% of existing materials mapped to new IDs
-  Zero data loss in migration

### ### Phase 2 (Compliance)

-  100% of EWC codes validated
-  Zero deals proposed violating T4 hazmat rules
-  Compliance check completes in <100ms
-  All existing waste streams have EWC codes within 30 days

### ### Phase 3 (Enhanced Agents)

-  ZOPA overlap detection accuracy >95%
-  Deal success rate increases by  $\geq 15\%$  (vs. baseline 3-round)
-  Average negotiation time <48h (from opening to proposal)
-  Zero negotiation deadlocks due to state machine bugs

### ### Phase 4 (Passports)

-  All new passports have complete composition data
-  Custody chain integrity = 100% (no broken hash chains)
-  Passport creation time <500ms
-   $\geq 30\%$  of passports upgraded to Q2+ within 90 days

### ### Phase 5 (Human Controls)

-  100% of HIGH risk flags reviewed by humans
-  Zero deals approved with blocking flags
-  Audit trail 100% complete (every action logged)
-  Emergency stop response time <2 seconds

### ### Phase 6 (Pricing)

-  Market prices updated daily (100% uptime)



- ☒ Price-based scoring reduces deal rejections by  $\geq 20\%$
- ☒ Agent learning improves acceptance rate by  $\geq 10\%$  after 50 deals

#### ### Phase 7 (UI)

- ☒ Material flow creation time reduced by 40% (vs. old UI)
- ☒ Passport viewer load time  $< 1.5s$
- ☒ User satisfaction score  $\geq 4.5/5$  (post-rollout survey)
- ☒ Zero UI regressions from old functionality

#### ### Phase 8 (Revenue)

- ☒ Payment processing 100% reliable (zero failed transactions)
- ☒ Subscription upgrades =  $\geq 15\%$  of Starter tier within 6 months
- ☒ Transaction fees collected on 100% of approved deals

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### ## \*\*Risk Mitigation\*\*

#### ### Risk 1: Material Taxonomy Complexity

**\*\*Impact:\*\*** 847 SKUs overwhelming for users, low adoption

**\*\*Mitigation:\*\***

- Smart defaults (suggest SKU based on industry)
- Guided wizard ("What form? → Shredded → Which grade? → No. 1")
- Search with fuzzy matching ("aluminum scrap" → 12 results)
- Most common 50 SKUs promoted in UI

#### ### Risk 2: Compliance Over-Blocking

**\*\*Impact:\*\*** Too many false positives, users frustrated by blocked deals

**\*\*Mitigation:\*\***

- Grace period (30 days) for EWC code backfill
- Override capability for admins (with mandatory reason)

- Soft flags vs. hard blocks clearly differentiated
- User education (webinars, docs) on compliance requirements

### ### Risk 3: Agent Negotiation Failures

**\*\*Impact:\*\*** New 5-round logic causes more deadlocks than 3-round

**\*\*Mitigation:\*\***

- A/B test (50/50 split) for 2 weeks before full rollout
- Rollback capability (feature flag)
- Monitor key metrics: deadlock rate, avg rounds to agreement
- Circuit breaker: if deadlock rate >25%, auto-rollback

### ### Risk 4: Database Migration Issues

**\*\*Impact:\*\*** Schema changes break existing queries, data loss

**\*\*Mitigation:\*\***

- Phased migration (add nullable fields first, backfill, then make required)
- Comprehensive backups before each migration
- Staging environment testing (100% prod data copy)
- Rollback scripts ready for all migrations

### ### Risk 5: Pricing API Costs

**\*\*Impact:\*\*** Real-time LME/ICIS APIs = €10k/mo, unsustainable pre-revenue

**\*\*Mitigation:\*\***

- Phase 1: Mock feeds (zero cost)
- Phase 2: Batch daily (not real-time) = 24x cost reduction
- Negotiate volume discounts with data providers
- Pass-through pricing for Enterprise tier users

### ### Risk 6: Performance Degradation

**\*\*Impact:\*\*** 847 SKU database + complex compliance checks slow down platform

**\*\*Mitigation:\*\***

- Indexed searches (material class, EWC code, keywords)
- Redis caching for material properties (24h TTL)

- Async compliance checks (don't block UI)
- Performance budgets: 200ms search, 500ms passport create, 100ms compliance check

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## ## \*\*Documentation Deliverables\*\*

### ### Developer Docs

1. **Material Taxonomy Guide** - How to add new SKUs, structure, naming conventions
2. **Compliance Integration** - EWC code selection, hazmat validation, cross-border rules
3. **Agent Architecture** - State machine, ZOPA calculation, negotiation protocol
4. **API Reference** - All new endpoints (material search, compliance check, passport CRUD)
5. **Database Schema Changelog** - All schema changes per phase

### ### User Docs

1. **Material Selection Guide** - How to find the right SKU for your waste stream
2. **EWC Code Reference** - Plain-language descriptions of common codes
3. **Digital Passport Tutorial** - Creating, viewing, transferring passports
4. **Agent Configuration** - Setting constraints, approval thresholds, emergency controls
5. **Compliance Best Practices** - Avoiding common pitfalls, required certifications

### ### Admin Docs

1. **Deployment Runbook** - Step-by-step rollout per phase
2. **Monitoring Dashboard** - Key metrics to watch, alert thresholds

3. **\*\*Rollback Procedures\*\*** - How to revert each phase if issues arise
4. **\*\*Compliance Audit Trail\*\*** - How to export for regulatory inspection

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## **\*\*Post-Implementation: Continuous Improvement\*\***

### **### Q1 2027: Phase 9 - Advanced Features**

- **\*\*Blockchain Integration:\*\*** Ethereum passport anchoring for immutability
- **\*\*AI Material Recognition:\*\*** OCR + computer vision to auto-classify waste from photos
- **\*\*Predictive Maintenance:\*\*** Agent recommends process changes to reduce waste generation
- **\*\*Carbon Credit Marketplace:\*\*** Verified emissions reductions → tradeable credits

### **### Q2 2027: Phase 10 - Geographic Expansion**

- **\*\*UK Compliance:\*\*** Post-Brexit waste regulations (different EWC codes)
- **\*\*US Expansion:\*\*** EPA waste codes, state-level regulations (50 variants)
- **\*\*Asia\*\*:** China GB, Japan industrial waste law, South Korea

### **### Q3 2027: Phase 11 - Industry Verticals**

- **\*\*Automotive-Specific:\*\*** IMDS integration, End-of-Life Vehicle Directive
- **\*\*Electronics-Specific:\*\*** WEEE fast-track, critical raw material prioritization
- **\*\*Construction-Specific:\*\*** Level(s) framework, demolition waste protocols

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## ## \*\*Budget Estimate\*\*

### ### Development Costs (15 weeks @ 2 FTE)

- Senior Full-Stack Developer:  $\text{€}85/\text{h} \times 40\text{h}/\text{wk} \times 15\text{wk} \times 2 \text{ FTE} = \text{**€}204,000\text{**}$
- QA Engineer (Part-Time):  $\text{€}65/\text{h} \times 20\text{h}/\text{wk} \times 15\text{wk} = \text{**€}39,000\text{**}$

### ### External Services

- LME/TSI/ICIS Test Accounts:  $\text{**€}1,500/\text{mo} \times 3\text{mo} = \text{€}4,500\text{**}$
- Compliance Consulting (EWC validation):  $\text{**€}8,000\text{**}$  (one-time)
- Legal Review (contract templates):  $\text{**€}5,000\text{**}$  (one-time)

### ### Infrastructure

- Supabase Pro (increased storage/compute):  $\text{**€}300/\text{mo} \times 3\text{mo} = \text{€}900\text{**}$
- Redis Cache (Upstash):  $\text{**€}150/\text{mo} \times 3\text{mo} = \text{€}450\text{**}$

### \*\*Total Phase 1-7: €261,850\*\*

### \*\*Total Phase 8 (Future): €45,000\*\* (subscription integration + Stripe)

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## ## \*\*Timeline Summary\*\*

- \*\*Week 1-3:\*\* Material Taxonomy + Compliance Framework
- \*\*Week 4-6:\*\* Enhanced Agent Logic + Negotiation
- \*\*Week 7-8:\*\* Digital Passport Enhancement
- \*\*Week 9-10:\*\* Human Control Layer
- \*\*Week 11-12:\*\* Pricing Mechanisms
- \*\*Week 13-14:\*\* UI Enhancements
- \*\*Week 15+:\*\* Platform Revenue (Future)

**\*\*Total Duration: 14 weeks (3.5 months) for Phases 1-7\*\***

**\*\*Production-Ready: Q2 2026\*\***

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## **## \*\*Next Steps\*\***

1. **\*\*Review & Approval:\*\*** Stakeholder sign-off on phasing + priorities
2. **\*\*Team Assembly:\*\*** Hire/assign 2 senior developers + 1 QA engineer
3. **\*\*Environment Setup:\*\*** Staging environment with full prod data copy
4. **\*\*Sprint 0 (Week 0):\*\*** Tooling setup, CI/CD pipeline, testing framework
5. **\*\*Sprint 1 (Week 1):\*\*** Begin Phase 1 - Ferrous metals taxonomy (68 SKUs)
6. **\*\*Daily Standups:\*\*** Track progress, identify blockers
7. **\*\*Weekly Demos:\*\*** Show progress to stakeholders, gather feedback
8. **\*\*End-of-Phase Reviews:\*\*** Verify success metrics before proceeding

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**\*\*This plan is ready for execution. All architectural decisions are explained, dependencies are mapped, and risks are identified with mitigations. The existing codebase provides a strong foundation – we're building on proven agent logic and deal-making infrastructure while adding the depth and compliance rigor needed for European market production readiness.\*\***